



Published in final edited form as:

Proc SPIE Int Soc Opt Eng. 2025 February ; 13405: . doi:10.1117/12.3047080.

Improving low-contrast liver metastasis detectability in deep-learning CT denoising using adaptive local fusion driven by total uncertainty and predictive mean

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Abstract

Emerging deep-learning-based CT denoising techniques have the potential to improve diagnostic image quality in low-dose CT exams. However, aggressive radiation dose reduction and the intrinsic uncertainty in convolutional neural network (CNN) outputs are detrimental to detecting critical lesions (e.g., liver metastases) in CNN-denoised images. To tackle these issues, we characterized CNN output distribution via total uncertainty (i.e., data + model uncertainties) and predictive mean. Local mean-uncertainty-ratio (MUR) was calculated to detect highly unreliable regions in the denoised images. A MUR-driven adaptive local fusion (ALF) process was developed to adaptively merge local predictive means with the original noisy images, thereby improving image robustness. This process was incorporated into a previously validated deep-learning model observer to quantify liver metastasis detectability, using area under localization receiver operating characteristic curve (LAUC) as the figure-of-merit. For proof-of-concept, the proposed method was established and validated for a ResNet-based CT denoising method. A recent patient abdominal CT dataset was used in validation, involving 3 lesion sizes (7, 9, and 11 mm), 3 lesion contrasts (15, 20, and 25 HU), and 3 dose levels (25%, 50%, and 100% dose). Visual inspection and quantitative analyses were conducted. Statistical significance was tested. Total uncertainty at lesions and liver background generally increased as radiation dose decreased. With fixed dose, lesion-wise MUR showed no dependency on lesion size or contrast, but exhibited large variance across lesion locations (MUR range ~0.7 to 19). Compared to original ResNet-based denoising, the MUR-driven ALF consistently improved lesion detectability in challenging conditions such as lower dose, smaller lesion size, or lower contrast (range of absolute gain in LAUC: 0.04 to 0.1; P-value 0.008). The proposed method has the potential to improve reliability of deep-learning CT denoising and enhance lesion detection.

Keywords

X-ray CT; deep learning; uncertainty; noise reduction; predictive mean

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1 INTRODUCTION

Using advanced CNN models, recent deep-learning-based methods have been shown to provide enhanced denoising capabilities for CT exams. However, like iterative reconstruction (IR) algorithms, deep-learning denoising at aggressively reduced dose can impair the detectability of challenging lesions, e.g., small and low-contrast liver lesions, which hampers further dose reduction in many clinical CT tasks. Moreover, the existing deep-learning methods typically generate point estimates of the underlying anatomical signal, i.e., denoised outputs are fixed for given inputs. Like many other machine-learning techniques, the point estimates are subject to two major uncertainties: *data uncertainty* (originating from noise / artifacts in inputs) and *model uncertainty* (originating from training-testing data dissimilarity) [1]: Data uncertainty could be quantified as the variance (or standard deviation) in CNN outputs for the same inputs across different noise realizations; Model uncertainty could be quantified as the variance (or standard deviation) in CNN outputs for the same inputs, but across different realizations of CNN model weights. At lower dose, these uncertainties could be amplified and further degrade the reliability of deep-learning denoising methods. Several recent studies have considered the quantification and / or the use of CNN uncertainty in clinical CT applications. Guo et. al. used noise propagation to estimate data uncertainty in a transparent CNN model (i.e., with known architecture and training data). [2] We combined knowledge distillation and Bayesian modeling to quantify patient-specific total uncertainty (i.e., data + model uncertainties) of non-transparent CNN models (i.e., unknown architecture and training data). [3] Also, we integrated explicit uncertainty and bias modeling to concurrently penalize uncertainty and bias in CNN-based material decomposition. [4] For CT denoising, Tivnan et. al. developed a loss function where sampling-derived data uncertainty was penalized during variational training. [5] For artifact reduction in cone-beam CT, Zhang et. al. merged the outputs of CNN and filtered-back-projection via model-uncertainty-based weighting maps. [6] However, few studies have fully exploited the characteristic information of CNN output distribution to enforce reliable lesion detection in CT denoising. In this work, we aimed to enhance low-contrast liver metastases detection in the CNN-denoised images by leveraging total uncertainty and predictive mean that, when combined, characterized CNN output distribution. Further details of this study are presented in the following sections.

2 METHODS

2.1 The proposed method

Data and model uncertainties in CNN-denoised images were separately quantified and then combined to derive total uncertainty and predictive mean (Figure 1). The mean-uncertainty-ratio (MUR) was calculated to quantify the dispersion of CNN outputs. If MUR was ≤ 1.0 , the CNN output distribution was highly dispersed and thus deemed highly unreliable. The corresponding CNN outputs were ignored. When MUR was > 1.0 , the corresponding predictive mean was viewed as more reliable compared to the original denoised images (i.e., point estimates). To mitigate the potential CNN bias, the predictive mean was further merged with IR images through an adaptive local fusion process. Finally, lesion detectability was quantitatively assessed by a previously validated deep-learning model observer (DLMO)

technique [7, 8] that can be directly applied to patient CT images. Further introduction of each step is presented in the following sub-sections.

2.1.1 Total uncertainty and predictive mean: For data uncertainty quantification, a set of noisy inputs with independent noise realizations could be acquired via repeated CT scans. This is impractical for patient CT exams. Therefore, a noise sampling process was developed to estimate the data uncertainty. Given well-trained CNN model, the difference images between CNN output and original noisy inputs mainly contained image noise. A random spatial decoupling process was applied to the difference images to approximate different noise realizations, involving random small angle rotation (angle range $[-5^\circ, 5^\circ]$, nearest-neighbor interpolation) and translation (in-plane shift range $[-3, 3]$ pixels) in the image domain. These noise realizations were added back to the original denoised images to create a set of noisy inputs which were processed by the same CNN (with fixed model weights). The corresponding CNN outputs were collected to quantify data uncertainty as pixel-wise variance. For model uncertainty quantification, Monte Carlo Dropout (MCD) [6] was applied to the target CNN model. This approach approximated the posterior distribution of CNN model weights. Briefly, a dropout operator was added after each convolutional layer and was activated during inference. For a given noisy input (with a fixed noise realization), the inference was repeated multiple times to collect the CNN outputs across random samples of CNN weights. Like data uncertainty, model uncertainty was also quantified as pixel-wise variance. To measure total uncertainty, the noise sampling process and MCD were conducted concurrently. Data uncertainty and model uncertainty were derived after applying marginalization across all samples of model weights and noise realizations, respectively. The total uncertainty was calculated as the sum of these two:

$$\sigma_{\text{total}}^2 = \sigma_{\text{data}}^2 + \sigma_{\text{model}}^2 \approx \frac{1}{T} \sum_{\omega_t} \sigma_{\omega_t}^2 + \frac{1}{T} \sum_{\omega_t} \mu_{\omega_t}^2 - \left(\frac{1}{T} \sum_{\omega_t} \mu_{\omega_t} \right)^2 \quad (1)$$

where $\sigma_{\omega_t}^2$ was the sample variance at t^{th} realization of CNN model weights ω across all noise realizations, and μ_{ω_t} the sample mean at t^{th} realization of CNN model weights ω across all noise realizations. The predictive mean was quantified as the mean CNN outputs across all realizations of noise and model weights.

2.1.2 Adaptive local fusion and DLMO: A scanning process was conducted, where a sliding ROI was used to assess local MUR. A weighting function used local MUR as its input to derive local weighting factors α and $1 - \alpha$ for local predictive mean images and the same ROI in noisy IR images, respectively (Figure 2). In this preliminary work, sigmoid function was used as the weighting function:

$$\alpha = \frac{1}{1 + \exp\left(-\frac{1}{\gamma_{\text{MUR}}}\right)} \quad (2)$$

where γ_{MUR} indicated the local MUR. The weighted sum of local predictive mean and IR images were used in the downstream analyses. A previously validated DLMO technique [7, 8] was applied to assess lesion detectability. Briefly, this DLMO was directly applied to patient images in lesion localization tasks; it was already well-calibrated to the mean performance of radiologist readers from prior work. To combine adaptive local fusion and DLMO, the scanning process above was performed simultaneous to the lesion searching process of the DLMO.

2.2 Experimental evaluations

The patient abdominal CT dataset from reference [7] was used to assess liver metastasis detectability. This dataset included 7 experimental conditions (Table 1), involving three lesion sizes (7, 9, and 11 mm), three lesion contrasts (15, 20, and 20 HU), and three radiation dose levels (25%, 50%, and 100% routine dose). Each condition included 100 trial cases (66 lesion-present, 34 lesion-absent). Each trial case comprised 5 consecutive images. Original patient images were reconstructed using commercial IR algorithm. For proof-of-concept, a ResNet-based CT denoising method [9] was implemented using paired routine and simulated 25% dose scans from an independent patient dataset. After training was completed, the ResNet model was used to denoise trial images in all conditions. Lesion-wise MUR was analyzed using boxplots. The previously-calibrated DLMO was deployed to evaluate the gain in lesion detectability before and after using MUR and adaptive local fusion. The area under localization receiver-operating-characteristic curve (LAUC) was used as the figure-of-merit of lesion detection performance. Statistical significance was tested using Wilcoxon signed rank test. P-value < 0.05 was deemed as statistically significant.

3 RESULTS

3.1 Predictive mean and total uncertainty

Examples of total uncertainty and predictive mean across different lesion size / contrast and radiation dose level are illustrated in Figure 3. The magnitude of total uncertainty increased as radiation dose decreased. With a fixed dose level, the spatial distribution and overall magnitude of total uncertainty varied across different liver background. Moreover, lesion-wise MUR yielded no apparent dependency on lesion size and contrast (Figure 4), while showing large dependency on dose level. This suggested the location and the corresponding liver background of lesions affected the total uncertainty per lesion in CNN outputs. In addition, lesion-wise MUR generally increased as dose increased, since data uncertainty decreased at higher dose.

3.2 Lesion detectability

Lesion detectability was close to maximum in the conditions with routine dose, larger lesion size or higher contrast (conditions #3 and #5) with and without using MUR and adaptive fusion. However, the proposed method consistently improved low-contrast liver metastasis detectability at lower radiation dose, smaller lesion size, or lower lesion contrast (Figure 5). The absolute gain of mean LAUC after using MUR and adaptive local fusion in the more challenging conditions (from conditions #1, #2, #4, #6 to #7) was 0.05, 0.10, 0.04, 0.06, and 0.09. The gain of lesion detectability was statistically significant (P-value 0.008).

4 CONCLUSIONS

An adaptive local fusion process driven by total uncertainty and predictive mean was developed to enhance low-contrast liver metastases detectability in patient images processed by deep-learning denoising. For proof-of-concept, the proposed method was demonstrated using patient abdominal CT exams processed by a ResNet-based denoising method. The change of lesion detectability in liver background was objectively quantified using a previously validated deep-learning model observer. The proposed method demonstrated the potential to further enhance lesion detection in this challenging CT task.

5 NEW WORK TO BE PRESENTED

The new work in this study is summarized as follows. The characteristic metrics of CNN output distribution (data uncertainty, model uncertainty, and predictive mean) were leveraged to enhance low-contrast liver metastases detection in patient images processed by CNN-based CT denoising method. Using these metrics, unreliable image regions were detected and excluded, while the remaining image regions were improved through a custom adaptive local fusion process. The proposed method achieved statistically significant improvement in lesion detectability, particularly in challenging experimental conditions with lower dose, smaller lesion size and / or lower contrast. Further experiments with additional CNN-based CT denoising methods and weighting functions for adaptive fusion are in progress.

ACKNOWLEDGEMENTS

Research reported in this work was supported by the National Institutes of Health under award number (R01 EB017095). The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institute of Health.

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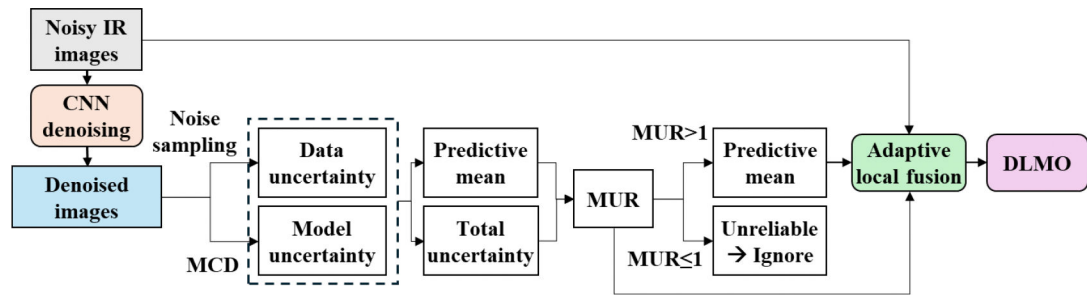


Figure 1.

The overall diagram of the proposed method. MUR – mean-uncertainty-ratio, DLMO – deep-learning model observer (DLMO), MCD – Monte Carlo Dropout.

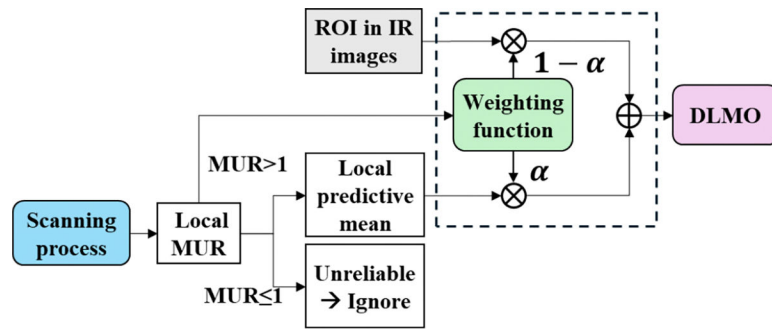


Figure 2.
The workflow of adaptive local fusion.

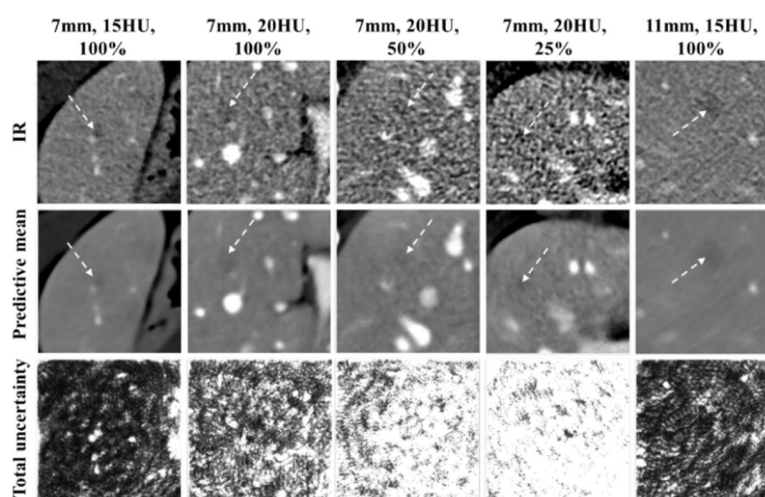


Figure 3. Examples of noisy IR images, predictive mean, and total uncertainty for 2 lesion sizes (7 & 11 mm), 2 contrasts (15 & 20 HU), and 3 dose levels (25% to 100%). Arrows indicate lesion locations

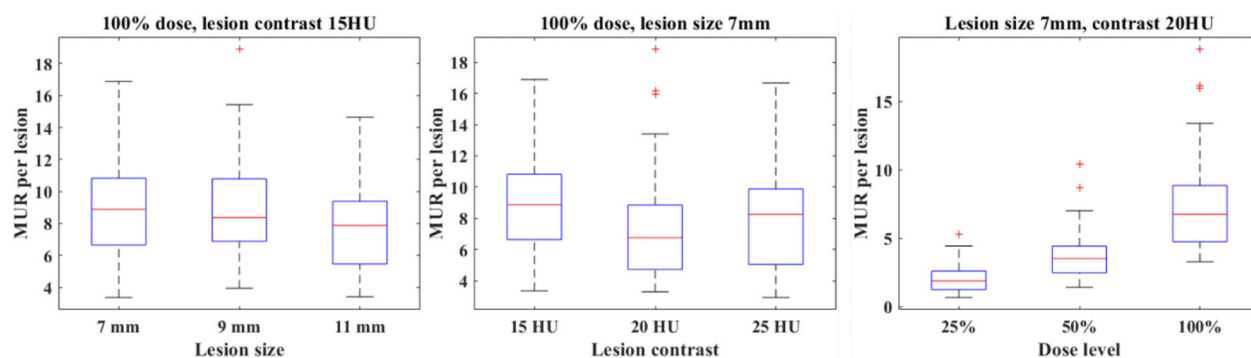


Figure 4.

Boxplots of lesion-wise MUR across different experimental conditions. MUR – mean-uncertainty-ratio. The presented MUR was calculated using small region-of-interests that centered at each lesion and tightly included immediate liver background.

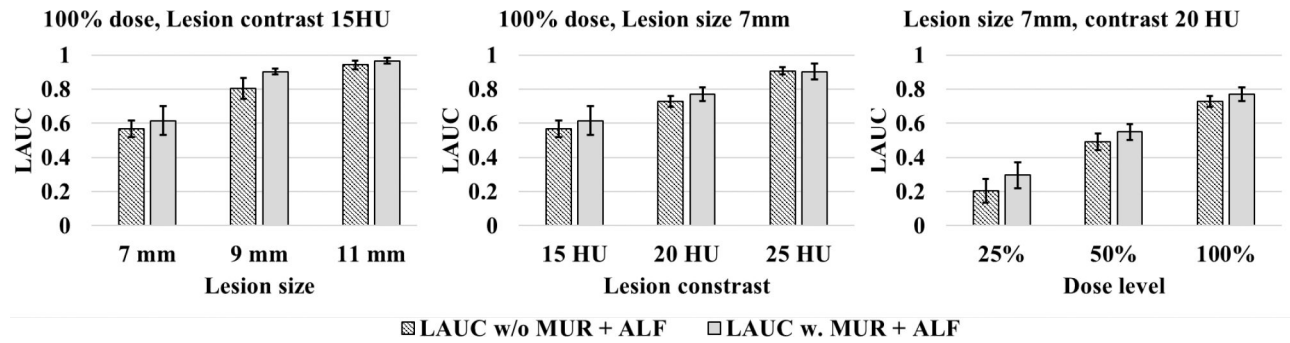


Figure 5.

Comparison of LAUC before and after using the proposed method across different experimental conditions. The length of error bars equal to twice of standard deviation.

LAUC – area under localization ROC curve. MUR – mean-uncertainty-ratio. ALF – adaptive local fusion.

Table 1

Experimental conditions involved in lesion detection *

Index	#1	#2	#3	#4	#5	#6	#7
Contrast (HU)	15	15	15	20	25	20	20
Size (mm)	7	9	11	7	7	7	7
Dose level	100%	100%	100%	100%	100%	50%	25%

* All images were originally reconstructed with commercial IR algorithm and then processed by ResNet-based demising method; Each condition included 100 trial cases